PakEmis 1.0.0rc69: technical account of greenhouse-gas and air-pollutant emissions in Pakistan, July 2015–June 2025

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Abstract. PakEmis reconstructs Pakistan's non-LULUCF greenhouse-gas and air-pollutant emissions from a shared national activity and material-flow account. Release 1.0.0rc69 covers ten fiscal years, July 2015–June 2025, with a separate source-limited provisional FY2025–26 extension. This technical report supports the national data-description article and maintained sector notes.

The S_TAN reference produces $444.6\,\mathrm{Mt}$ CO_2-equivalent in FY2023–24 and $457.2\,\mathrm{Mt}$ in FY2024–25 on AR5 GWP-100. Native CO_2, CH_4 and N_2O remain separately reported. Reference-year covered emissions include $422.8\,\mathrm{Gg}$ primary $PM_{2.5}$, $816.6\,\mathrm{Gg}$ NO_2-equivalent NO_x and $2670\,\mathrm{Gg}$ NH_3. Seven core air species have substantive source methods, results and evaluation; additional particulate strata retain narrower coverage.

Station coal consumption and rank-specific NCVs, reconciled consumer accounts, survey-supported household fuels and conserved nitrogen pathways distinguish this reconstruction from a clipped global inventory. Shared conditional realisations produce native/GWP levels and paired trend intervals, with structural allocations, control scenarios and missing sources kept separate. Monthly and geographic products retain process-specific support, unallocated mass and licensing boundaries. Exact-period public HTAP/EDGAR comparisons identify shared inputs and important source discrepancies. The locally prepared review package includes data, code, metadata, figures, reproduction instructions and checksums; external review deposit and a persistent identifier remain pending. The dataset DOI is assigned on deposit; until then the release is identified by version and manifest digest.

1 Introduction

Pakistan's emission sources are represented in international inventories, but national activity, material transfers, reporting boundaries and source geography are not interchangeable with their global defaults. Public regulators and statistical agencies provide fuel, production, crop, nutrient, household and demographic records at different periods and administrative scales. Reconstructing compatible activities and documenting revisions makes those sources usable together.

PakEmis provides one greenhouse-gas and air-pollutant inventory based on that reconstruction. The common core covers July 2015–June 2025. Annual source records, compatible monthly estimates, administrative allocations, native cell masses, uncertainty and query products describe the same named configurations. The reference is S_TAN: survey-supported household dung and a coherent total-ammoniacal-nitrogen formulation. L and the IPCC nitrogen alternative are retained separately.

The contribution is source evidence and accounting rather than a larger species list or finer nominal grid. Public-power coal combines station consumption and rank NCVs rather than treating the publisher's aggregate energy conversion as an unquestioned physical account. Industrial coal retains unresolved quantities and dated kiln alternatives. Household surveys identify supported combustion while leaving other diverted dung destinations unresolved. Nitrogen pathways link NH_3, NO_x and N_2O from the same manure and soil states.

The core air panel is primary $PM_{2.5}$, BC, NO_x, SO_2, CO, NMVOC and NH_3. Their coverage, control assumptions and emission boundaries differ by process. Supplemental PM_{10}, TSP and OC are retained where supported. BC/OC are particulate components; air mass is not CO_2-equivalent and is not ambient concentration or exposure. No LULUCF, chemistry model or health assessment is incorporated into the core.

One national data-description article is supported by this expanded technical report and maintained sector notes. Quantitative claims here use current released products. Historical debugging and numerical revisions belong in the release change record; the following sections describe the final methods, limitations and useful comparisons.

1.1 Related inventories and methodological context

National reporting follows IPCC category, native-gas and metric conventions (IPCC, 2006, 2019). EDGAR provides internally consistent international series but uses energy balances and other inputs that may be shared with the Pakistani public sources (Crippa et al., 2025, 2024). Agreement with those products is therefore not automatically independent validation. Calendar and fiscal periods, domestic/international transport, biomass and fossil fuels, process boundaries and missing categories must be matched before comparing totals.

CAMS-GLOB-ANT, HTAPv3.2 and MIXv2 demonstrate source/species coverage tables, temporal allocation, versioning and evaluation appropriate to inventory data descriptions. Their presentation and comparison methods inform this release; they are not proof of its accuracy. Primary articles are available at https://essd.copernicus.org/articles/16/2261/2024/, https://essd.copernicus.org/articles/17/5915/2025/ and https://acp.copernicus.org/articles/24/3925/2024/.

The air methods use applicable IPCC, EMEP/EEA and AP-42 source factors, with Pakistan-specific activity and explicit technology/control transfers. National contributions arise where local station records, consumer accounts, household surveys, nutrient/product histories and facility evidence alter or explain a global-inventory result. Coarse information downscaled to a fine grid retains its original support, and a good fuel-carbon balance does not validate transferred particulate or NO_x factors.

2 Scope and shared source framework

The dataset is a Pakistan-specific non-LULUCF reconstruction of supported greenhouse gases and primary air pollutants over July 2015–June 2025. The four integrated configurations cross household dung S/L and nitrogen TAN/IPCC choices. Source status, period and configuration accompany every product; a provisional FY2026 source extension is separate from the common core.

2.1 Scope and canonical source records

Each source record distinguishes physical source, emitting process, reporting category, calculation module, native species and chemical mass convention. Activity quantity, unit, fiscal dates and provenance accompany factor applicability, technology, control assumptions, configuration, estimate maturity, coverage and temporal/spatial support. Calculation-module totals are implementation summaries; category accounts determine scientific reporting and prevent captive fuel or process carbon from being counted twice.

For an appropriate activity/factor process,

$$E_{s,p,t} = A_{s,t} EF_{s,p,t} (1 - \eta_{s,p,t}). \tag{1}$$

This expression is not applied indiscriminately. Sulfur balances, surviving nitrogen pools, material conversion and recovered process carbon have their own equations. A factor that already describes controlled operation is not discounted again. Activity is shared where the physical stream is shared; pollutant-specific chemistry and process boundaries remain distinct.

2.2 Fuel and industrial activity

OCAC product-specific sales are reconciled at compatible use boundaries before road, off-road, industrial, railway and household fuel are partitioned. Overlapping editions identify historical revisions; a revised national total does not invent unpublished revised sector cells. OGRA deliveries, physical production and infrastructure distinguish gas use from fugitive pathways. Unaccounted-for gas is not interpreted wholly as methane leakage.

Public-power coal uses station consumption times rank-specific NCV. Generation and heat rates provide checks or explicit alternatives, not another input on top of the consumed fuel. K-Electric membership is tested fuel by fuel. Industrial coal distinguishes cement integrated kilns, brick firing, other industrial uses, captive fuel and unresolved supply/stock/boundary quantities. Dated fleet observations do not constitute an annual operating census.

IPPU uses source-specific physical output and trade with the correct process boundary. Cement/clinker production and trade, ammonia/feedstock and recovered carbon, steel and soda ash remain distinct from their combustion fuel. An aggregate manufacturing index does not supply missing physical clinker, ammonia or steel output. Missing process/control records remain coverage statuses.

2.3 Household, agriculture and waste activity

Household quantities combine deliveries and HIES survey evidence with explicit interpolation/extrapolation. The S dung reference is observed household use reconstructed between survey rounds; L assigns all ledger-diverted dung to household combustion. Dry solids including ash, volatile solids, as-received fuel and fuel energy are distinct quantities.

Crop area/production and fertiliser nutrients use compatible fiscal controls. Livestock is reconstructed consistently around the adopted census rather than spliced from an older projected baseline. The nitrogen ledger follows management, grazing and applied pools. Fertiliser offtake is distinct from application and release timing.

Waste advances the disposal-history state with demographic activity and the supported treatment/fate formulation. Fixed management parameters are not annual management observations. Waste origin, disposal, treatment and burning location are distinct supports. Aviation uses domestic traffic and fleet/LTO assumptions separately from fuel uplift, and railway traction/non-traction remains explicit.

2.4 Coverage and notation statuses

The inventory remains non-LULUCF. Supported CO_2, CH_4 and N_2O are reported as native gas masses, with GWP aggregates separately. The core air panel comprises primary $PM_{2.5}$, BC, NO_x, SO_2, CO, NMVOC and NH_3. PM_{10}, TSP and OC retain their narrower source coverage. Estimated module cells do not establish a complete national source account.

Status	Meaning
E	Supported estimate with source/process, period, configuration and maturity recorded
NE	Known source/species not estimated; reason retained, never filled with zero
IE	Included elsewhere under an explicit category or transfer boundary
NO	Source reported not occurring where the evidence supports that statement
NA	Species/process combination not applicable under the stated method

Table 1. Notation keys used in the current release. Source–process coverage is more detailed than the module summary.

2.4.1 Supported processes and material gaps

Industrial combustion includes admitted gas, liquids, cement/kiln coal, brick firing and captive gas. The coal account distinguishes physical station power, industrial uses and unresolved quantities. An external industrial total minus this estimate is a signed comparison, not a lower or upper bound on missing emissions. Source revisions and allocation choices can change that difference without changing source coverage.

IPPU includes cement process CO_2 and supported ammonia, steel and soda-ash process estimates at their admitted boundaries. A soda-ash category is not NA merely because an earlier release lacked a method. Fertiliser product tonnage is not ammonia output, and product/output, feedstock carbon and recovered CO_2 remain separate. Nitric acid, lime, glass, urea-finishing air releases and other unresolved industrial sources retain NE statuses where their physical process/control evidence is insufficient.

F-gases, coal-mining fugitive coverage and remaining waste-treatment sources are screened separately where supported. No global inventory estimate is added to the central total to close a missing category. Commercial/institutional fuel use and domestic navigation retain their documented boundaries or omissions. LULUCF exploratory material is excluded from every national core total and comparison.

Power air admission covers reviewed oil/diesel SO_2. Coal, gas and bagasse pollutant/control cases remain separate screening or design scenarios. Public licences establish proposed equipment at selected coal stations but not annual operation. Other

central power species therefore remain NE. Industry, IPPU and fugitives have admitted provisional air estimates; they are not absent simply because older aggregate descriptions omitted them.

Road air includes Tier-1 hot/cold exhaust, evaporative NMVOC, tyre/brake and road-surface wear. Off-road machinery and resuspended road dust are separate missing air estimates. Household fossil and biomass emissions are included at the gross appliance boundary; generators remain an air gap. Agricultural NH_3 and NO_x use the shared nitrogen pathways, while machinery and tillage/harvest dust remain missing sources. Waste burning is estimated with origin-based geography; treatment/incineration estimates are not manufactured from waste generation.

2.4.2 Transfers and non-additive quantities

Household electricity is counted where generation occurs. Captive self-generation reports under industrial combustion. Diverted dung and crop residues leave their source ledgers once; reference S receives observed household fuel, and unresolved dung destinations are not automatically burned. L is an explicit alternative destination assumption. Biogenic CO_2 remains a memo item under its documented convention.

BC and OC are components of $PM_{2.5}$, and $PM_{2.5}$ is contained in compatible PM_{10}/TSP. Their masses are not added across nested species. NO_x is NO_2-equivalent and NH_3 is molecular NH_3, distinct from intermediate nitrogen masses. No ozone or total ambient particulate is estimated by summing precursor emissions.

3 Source data and processing

The source hierarchy favours compatible national observations and held/public primary documents. Publication title alone does not establish complete fiscal coverage. The source availability record identifies edition, exact months, revision status, units, variables, location and actual usability by sector/year. Transcribed records retain source table/page or worksheet identity; raw public bytes and retrieval hashes preserve the evidence.

Source family	Principal core use and limitation
NEPRA, CPPA and K-Electric	Physical station fuel, generation, status and boundaries; public controls are not consignment fuel analysis
OCAC and OGRA	Product sales, deliveries, production and infrastructure; later overlap revisions and feed/-fuel boundaries retained
PBS QIM/minerals	Physical item/month/province controls where the source supplies them; aggregate indices do not provide missing process output
Economic Survey, PBS census and NFDC	Fiscal crops, nutrient/product offtake and census-anchored herd reconstruction; application timing and annual herd enumeration not observed
HIES and demographic records	Household fuel and population histories; survey interpolation/extrapolation and waste management assumptions labelled
PCAA and Pakistan Railways	Domestic/international traffic, fuel and operator controls; traction, LTO/cruise and fuel-uplift boundaries separate

Table 2. Activity sources and the distinctions needed for their scientific use. Exact vintages, months and controls are in the availability/source registers.

The latest core uses FY2024–25 supportable annual controls. FY2025–26 is investigated source by source: full-year OCAC records are distinguished from July–March Economic Survey fuel figures, and supportable QIM/source extensions retain their own scope. A partial sum never becomes a common national total. Overlapping editions identify historical revisions without manufacturing revised category cells.

PBS grouping preserves vertically centred mineral labels, province continuations, units and printed year blocks. Worksheet selection uses schema/provenance rather than equality of legitimate values. Final exported province/month records reconcile to compatible controls, with source anomalies and exclusions retained. Missing provinces are neither invented nor zero-filled. Profile evidence is downgraded or replaced where complete national support cannot be established.

Sources and factor/control evidence are identified in machine-readable registers. Public documents can be used as evidence without presuming they may be republished under this project's licence. Source retrieval, compiled-input regeneration and primary-source re-extraction are distinct reproduction stages.

4 Dataset generation

The dataset is generated in four stages that this section follows in order. The annual reconstruction computes every category–gas cell for the reference configuration and for each fiscal year of the panel, and closes it exactly to the national total. The monthly estimate resolves each annual cell into twelve months on a declared temporal profile, and records for every stream whether that profile was observed, borrowed or flat. The spatial allocation places the national mass on administrative units and on a regular grid, and keeps the part it cannot place as an explicit unallocated remainder, so that the closure identity is raster plus unallocated rather than raster alone. The uncertainty stage carries parameter-level draws through the same engine and prices the structural alternatives that a parametric interval cannot contain. The ledgers between sectors—the quantities estimated once and consumed twice across an accounting boundary—are part of the generation rather than of its validation, because they are what makes the total a single sum instead of a collection of sector totals.

4.1 Native gases and GWP aggregates

Reference S_TAN produces $444.6\,\mathrm{Mt}$ CO_2-equivalent in FY2023–24 and $457.2\,\mathrm{Mt}$ in FY2024–25 on AR5 GWP-100. Reference native masses are $161\,\mathrm{Mt}$ CO_2, $8621\,\mathrm{Gg}$ CH_4 and $153.4\,\mathrm{Gg}$ N_2O. AR6 aggregates are $437.8\,\mathrm{Mt}$ and $450.2\,\mathrm{Mt}$, respectively. The supported scope remains non-LULUCF and excludes explicit missing categories.

Household combustion supplies approximately two-thirds of reference covered $PM_{2.5}$; open waste and field burning are substantial additional sources. Nitrogen pathways dominate NH_3. NO_x and SO_2 retain different fuel, industrial and control contributions. These are specific source comparisons, not an inverse ordering of GHG and air emissions or an exposure ranking.

Native core species (Gg)	FY2023–24	FY2024–25
$PM_{2.5}$	422.8	422.7
BC	40.44	46.64
NO_x (NO_2-equivalent)	816.6	833.1
SO_2	213.6	368.9
CO	5991	5999
NMVOC	1187	1226
NH_3	2670	2690

Table 3. Covered-source air estimates from the current release. Nested particulate species are non-additive.

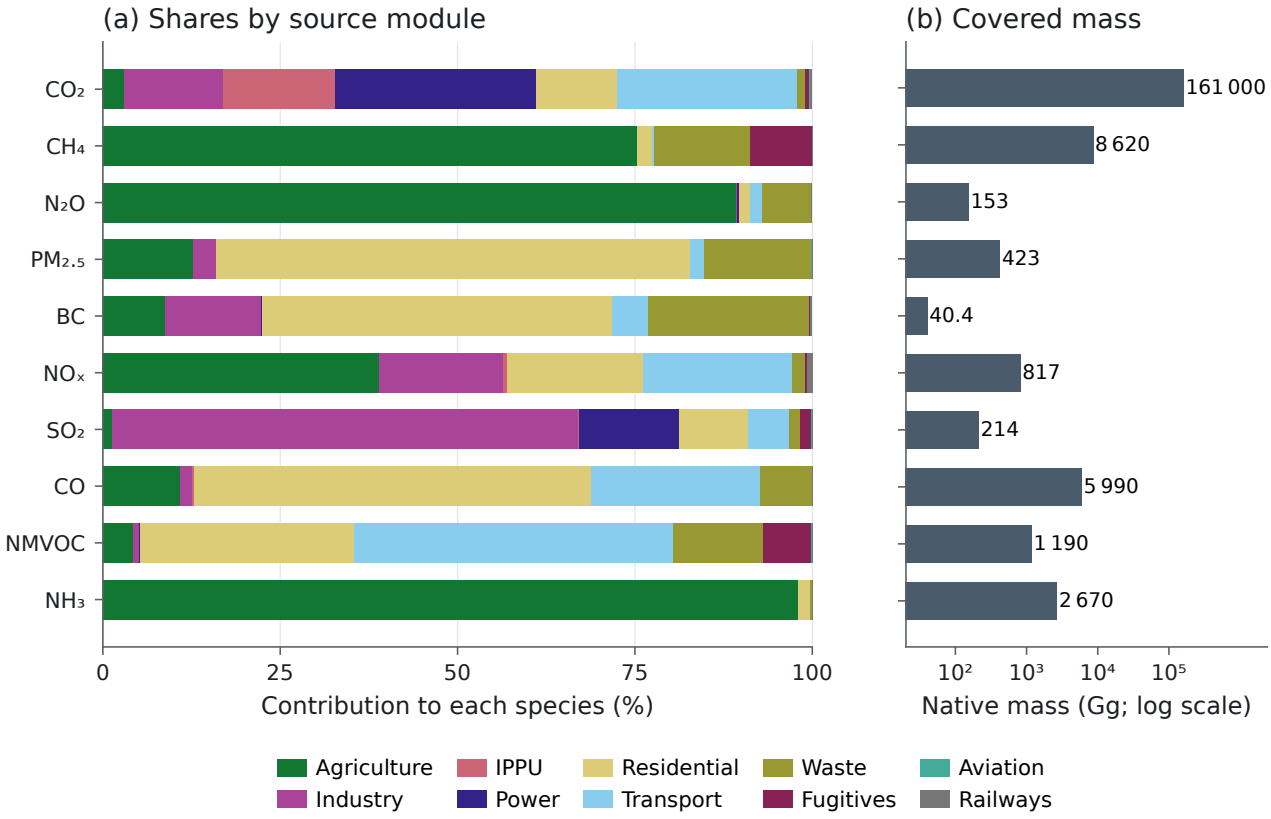


Figure 1. Reference source contributions within each species and separate native mass. Shares and absolute quantities answer different questions; the mass axis is logarithmic. Data accompany the figure.

4.2 Historical change

The common panel contains ten fiscal years, July 2015–June 2025. Between FY2015–16 and FY2024–25, covered AR5 GHG changes by $14.9\,\%$, $PM_{2.5}$ by $24.0\,\%$, NO_x by $18.2\,\%$, SO_2 by $-45.0\,\%$ and NH_3 by $45.7\,\%$. The provisional FY2025–26 source extension is excluded from these endpoints.

Station consumption, sales, crop and nutrient observations supply measured activity changes where available. Census-anchored herd reconstruction, household interpolation/extrapolation and fixed fate parameters supply modelled changes. Coal consumer allocations and edition revisions can cause important industrial changes; a shift in booked kiln fuel is not evidence of a regulatory change in district emissions. Waste history is advanced consistently. Older 1990 calendar diagnostics remain separate and are not spliced onto the common core as uniform fiscal observations.

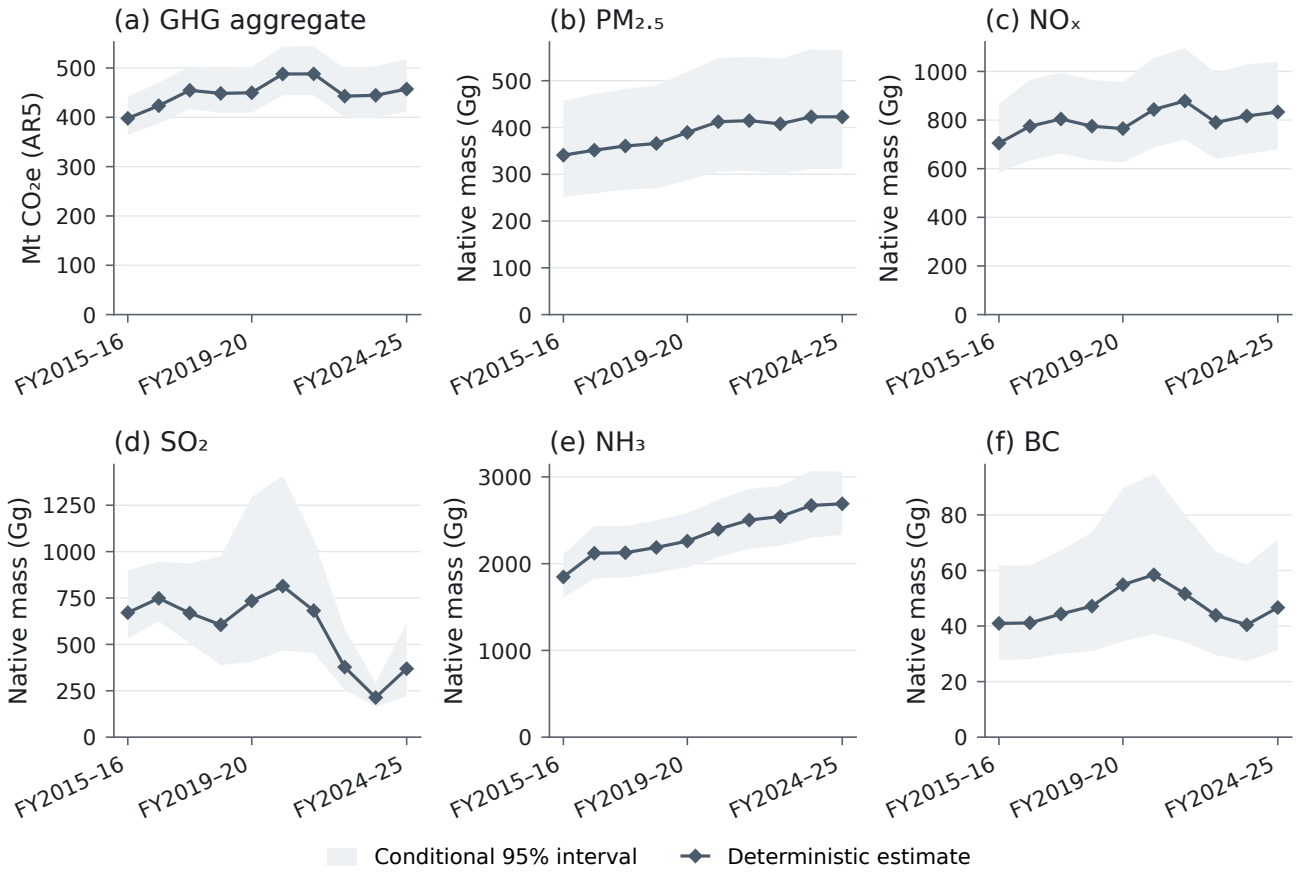


Figure 2. Fiscal trends and conditional persistent-parameter intervals. Structural alternatives and missing categories remain outside these intervals. The common core ends in June 2025.

4.3 Monthly allocation

Annual compatible estimates are allocated through source/process profiles,

$$E_{s,p,m} = E_{s,p,t} w_{s,p,m}, \qquad \sum_{m \in t} w_{s,p,m} = 1. \tag{2}$$

Production or sales can be a proxy for emission timing. Fertiliser offtake is supply timing rather than observed application; manure-stage and residue-decomposition release dates are unobserved. Road evaporation and wear do not inherit an observed exhaust fuel profile without evidence. Waste burning, annual flare quantities and unsupported releases retain declared fallbacks. The admitted station subset's aggregate fleet timing is labelled a proxy.

The repaired PBS extraction preserves centred labels, continuation/province rows, source units and worksheet identity. Actual exported province/month values and compatible monthly/annual controls determine admission. Surviving source anomalies are flagged, and national monthly classification requires producing-area completeness; missing observations are not converted to zero. Temporal evidence shares are calculated separately by pollutant.

4.4 Administrative and grid products

Source-matched annual weights produce ADM1, ADM2 and 0.05° cell masses under each configuration. Facilities respect commissioning/closure; selected oil/diesel stations use their admitted subset frame, and kiln types retain the supported frame plus an unlocated remainder. Rail geometry excluded from the compatible primary package retains supported administrative mass and explicit fine-scale unallocated mass. Fixed proxies and reference-year patterns retain their evidence class.

For each species/configuration/period, annual–monthly and placed–unallocated closure is tested at compatible boundaries. Cell mass is distinct from flux density. Area, country fraction where relevant and duration are required for conversion; vertical injection and detailed chemical speciation remain additional atmospheric-model requirements.

4.5 Shared accounting and material constraints

Shared accounts prevent duplicated source inputs, while explicit residuals retain unresolved evidence. Their numerical closure does not prove the split of material among destinations. Transfers, physical balances and publisher residual identities are different operations.

4.5.1 Dung and residue destinations

Agricultural nitrogen and volatile-solid rates determine diverted dung supply. Reference S receives household survey-supported fuel; material without a supported household destination remains unresolved and is not silently burned. L is the separate all-diverted-household case. The same configuration is applied to GHG, air, monthly, administrative and grid products. Crop residue supply is partitioned among returned soil material, field burning, household fuel and unresolved uses. Household and field combustion are distinct processes and geography.

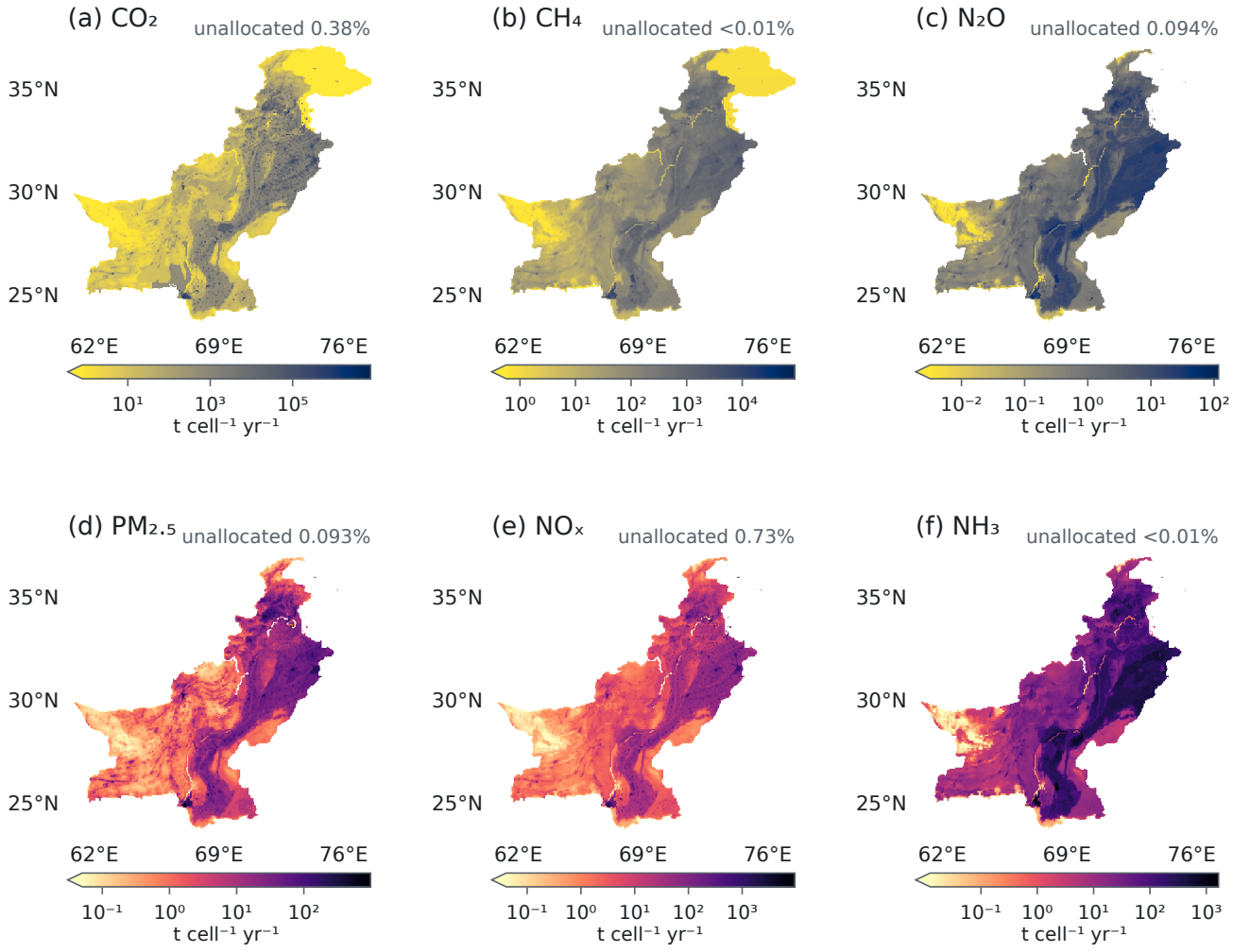


Figure 3. Reference native gas and pollutant mass per cell, with separate logarithmic scales. Geographic weights can describe release location, activity origin or accounting; fixed allocation is not observed local change. Unallocated mass is published separately.

4.5.2 Liquid fuel partitions

Household generator petrol and railway diesel are carved out of compatible sales controls before road fuel is estimated. On-road and off-road activity retain their category distinctions. Legitimate records with equal values are not deduplicated by value. Class/fuel/equipment partitions sum to the shared control and are not varied independently as unconstrained uncertainty multipliers.

4.5.3 Waste fate

Collected/disposed, uncollected and burned quantities preserve the supported waste-fate constraints. Disposal CH_4 advances the history state rather than using one current year's population as the entire disposal stock. Native uncertainty reruns preserve the material partition and supply the same burned dry mass to air factors. Waste emissions at origin do not establish the physical treatment location.

4.5.4 Nitrogen pathways

Reference TAN follows species/system NH_3 losses through housing or yards, storage and field application, acting on surviving total ammoniacal nitrogen. Managed NO_x, leaching, N_2 and N_2O are retained in the common state. Grazing and soil volatilisation remain distinct. The air NH_3/NO_x and GHG N_2O use the same losses and surviving pools; a separate IPCC configuration regenerates both product families.

Synthetic product factors are kg NH_3-N per kg N and receive the 17/14 molecular conversion once. Soil NO_x is calculated on nitrogen and reported as NO_2-equivalent using 46/14. The combined volatilised nitrogen used for indirect N_2O includes the applicable NH_3 and NO_x pathways rather than applying an extra ammonia share to a factor already in NH_3-N.

4.5.5 Coal and gas boundaries

The coal account records production, imports, exports/stock information where supported, station power and industrial consumers. Publisher consumer residuals do not establish independently measured use. Unresolved quantities remain explicit; they are not assigned wholesale to kilns. Physical fuel, rank heat and carbon factors remain separate in uncertainty draws. Delivered gas, process feedstock, combustion, leakage, flaring and venting preserve their boundaries without interpreting every flow difference as emissions.

4.6 Conditional uncertainty and structural alternatives

4.6.1 Parameter ensembles

The annual ensembles use 10 000 common parameter realisations with recorded seed and named dependence setting. Positive activity/factor multipliers use their declared distributions; physical fractions and native fate parameters retain bounds and

material constraints. Shared activity is represented by actual common draws rather than by summing independent sector output errors.

Nitrogen replays native NH_3, NO_x and N_2O from the same herd, crops, nutrient inputs and surviving state. Synthetic fertiliser product factors share their applicable input realisations with indirect N_2O. Waste fate regenerates the burned dry mass supplied to air factors, without resampling population or collection a second time. Fugitive activity/control realisations remain distinct from unrelated infrastructure measures. Ledger-L household dung uses the same sampled herd as manure accounting. Coal rank heat and carbon factors preserve the station reference.

Remaining source factor/technology assumptions are conditional where their dependence or performance is not observed. Equipment and manure-system shares are held rather than independently multiplied outside their partition. Particulate containment is checked at matched central source/process boundaries; the marginal factor transfer model does not constitute a measured joint particulate composition distribution. Sulfur policy/specification and plant operating alternatives are structural evidence, not sampling distributions merely because endpoints are available.

The reference national AR5 parameter interval is $399.9\,\mathrm{Mt}$ to $503.2\,\mathrm{Mt}$ CO_2-equivalent.Covered air intervals are $311.2\,\mathrm{Gg}$ to $567\,\mathrm{Gg}$ $PM_{2.5}$, $660.6\,\mathrm{Gg}$ to $1029\,\mathrm{Gg}$ NO_x and $2297\,\mathrm{Gg}$ to $3069\,\mathrm{Gg}$ NH_3.These are conditional parameter percentiles under S_TAN and persistent parameters. Missing categories, structural coverage/control choices and temporal/spatial allocation uncertainty are not incorporated.

4.6.2 Paired trends

Each change is calculated within a paired realisation before percentiles are taken. Persistent-parameter and annual-activity-error cases test unidentified temporal dependence. Subtracting unrelated annual percentiles does not produce a trend interval. Source revisions, maturity and coverage are retained in the interpretation, so a partial FY2026 sum never enters a common national change.

4.6.3 Structural cases

The four integrated configurations are S_TAN, L_TAN, S_IPCC and L_IPCC. Household S/L and nitrogen formulations are separate structural axes and are applied consistently to both product families. Other source cases include coal allocation/operating season, sulfur evidence, transferred road factors, household technology/fuel choices, waste burning fraction and regional fertiliser factors. Scenario values have no assigned probabilities and their extrema are not 95% limits.

Dated kiln-fleet cases keep observed dates, fleet coverage and season assumptions. Cases with a negative supply-minus-use quantity expose incompatibility rather than force closure; they are not mixed into an interval simply to cover an external total. Selected-plant FGD design cases apply an assumed efficiency once, without claiming demonstrated operation or an upper bound.

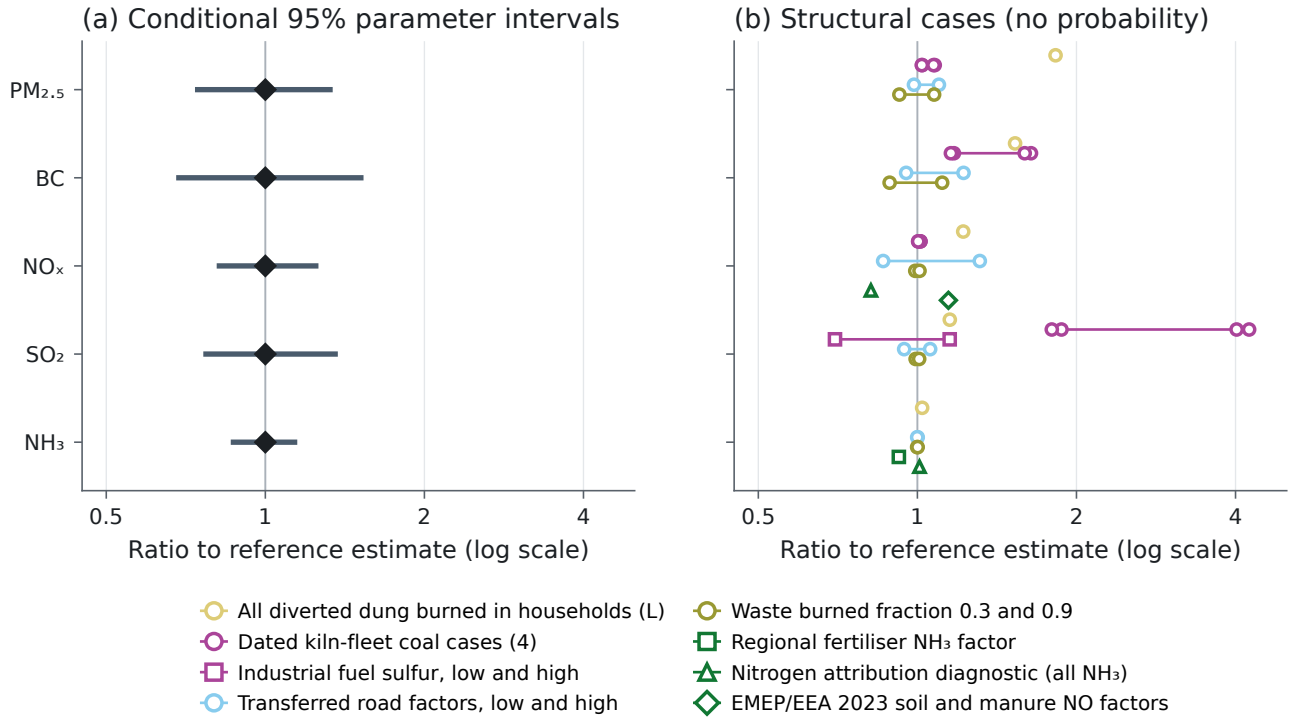


Figure 4. Conditional air parameter intervals and consequential structural cases relative to the reference. Structural points have no probability and are not confidence bounds.

4.6.4 Dominant uncertainty and reporting

Pollutant-specific driver records distinguish incomplete-combustion factors and household/waste activity from nitrogen and sulfur inputs. Correlation sensitivity diagnostics are not additive variance shares. Managed-soil factor applicability remains separate for irrigated cropland, rice and grazing; agreement with FAOSTAT does not select the reference factor. Unestimated sources and allocation uncertainty remain explicit limitations even where a parameter interval is relatively narrow.

5 Data records

Annual source atoms retain native species, source/process, category, configuration, reporting period, units, provenance, maturity and coverage. Rollups compose the four configurations consistently. Monthly records carry process-specific profiles/evidence. ADM1, ADM2 and annual/monthly cell products publish placed mass and an explicit unallocated ledger. Conditional levels/trends, driver/parameter records and structural cases accompany the central quantities.

Family	Fitness for use
Native annual and GWP summaries	Supported source/category totals; native gas mass distinct from metric aggregate
Core air annual/monthly	Covered source emissions in native Gg, with process/species coverage and evidence
Administrative and cell mass	Allocations under documented geometry and reporting masks; placed plus unallocated closure
Conditional uncertainty/trends	Parameter percentiles within a named configuration and paired dependence case
Structural/provisional products	Separate source/control/fate cases or exact partial-period extensions; no probability or uniform maturity implied
Registers and query/downloads	Source/factor/control provenance, periods/configurations, evidence, product licences and checksums

Table 4. Current product families. The machine-readable catalogue gives exact names, schema and advertised files.

The source/species matrix distinguishes E, NE, IE, NO and NA; unknown coverage is not a numerical zero. The fourteen-species vocabulary does not imply fourteen complete national inventories. Nested particulate species remain non-additive, NO_x uses its stated NO_2 convention and NH_3 differs from intermediate NH_3-N.

Reference-year sector records sit beside the national products under `outputs/`. For industrial fuel combustion, `industry_totals_2024.csv` gives the category totals, `industry_by_stream_2024.csv` resolves them to fuel streams and gases, `industry_provincial_2024.csv` allocates them to provinces and `industry_monthly_2024.csv` gives the twelve-month layer. `national_monthly_2024.csv` is the reference-year monthly layer across sectors, `air_pollutants_provincial_2024.csv` the provincial air-pollutant allocation by source and proxy, and `residential_configurations_by_year.csv` the dung mass, its activity basis and dung-stream $PM_{2.5}$ for each fiscal year and residential configuration.

CSV and query/download interfaces use the same current period/configuration vocabulary. A historical alias resolves to an explicitly named current configuration or retains a separate historical product identity; it does not select an unrelated file silently. Figures retain plotted data and native units. Exact file existence, openability and checksums are verified in the release preparation record.

6 Technical evaluation

Internal reconciliation verifies extraction, units, material constraints, aggregation and product interfaces. External evidence evaluates magnitude, historical change, timing and location with matched species/period/boundary and shared inputs recorded. These different checks do not imply the same level of validation. Internal source/export, material-flow, annual/monthly and placed/unallocated checks establish consistency. External evidence evaluates magnitude, change, timing or location at matched boundaries. The comparison records retain species, units, periods, provider and shared inputs; national agreement is not sufficient to select a factor or validate spatial placement.

6.1 Magnitude and category boundaries

National reporting and EDGAR are compared by native gas, GWP convention and source category. Fiscal and calendar periods are retained explicitly. Energy-balance inputs shared with Pakistani statistics limit independence. A signed industrial difference is not missing emissions and can reflect revised coal allocation, calorific basis, controls or coverage.

Industrial air comparisons distinguish EDGAR fossil fuel from fossil-plus-biomass and include matched cement process emissions where the PakEmis integrated class requires that boundary. Other IPPU remains separate. The latest fully supported fiscal comparison is FY2021–22 because the air comparator ends in calendar 2022. Later PakEmis years are not compared using extrapolated external values. Large BC/CO differences motivate factor/equipment investigation rather than tuning the local account to the comparator.

HTAPv3.2 provides public annual/monthly country bundles without credentials. Original Pakistan NH_3 rows are reconciled against printed annual controls before exact July–June matching. Agriculture and road comparisons cover FY2015–16 through FY2019–20; later complete fiscal external values are absent because the source ends in calendar 2020. Household gross emissions and waste burning differ from the external broader boundaries and therefore receive contextual values, without a misleading ratio.

HTAP uses REAS regional baselines, EDGAR gap fills/agricultural burning and EDGAR post-2015 trends. Shared statistics and provider provenance limit independence; agreement with both HTAP and EDGAR is not two independent constraints. FY2019–20 S_TAN agricultural NH_3 is approximately 2215 Gg against 2184 Gg in HTAP. Component pathways and factor applicability remain important even when aggregate magnitudes are close.

6.2 Change and timing

Overlapping source editions separate historical revisions, source additions and actual activity change. Monthly source rows and compatible national/annual controls are checked on the final exported panel. Uniform duplication can cancel in a normalised profile but still produces a false activity export; it is repaired at source. Correct extraction does not eliminate publication anomalies, which retain downgraded evidence and declared fallbacks.

Seasonal source evidence evaluates relevant emitting processes. Sales, crop calendars, fertiliser supply and fire detections are proxies with different physical meanings. An unsupported application or evaporation profile is not called observed solely because it spans twelve months.

6.3 Location

Selected facilities, administrative activity and held-out district crop records test location. Inputs used to construct a proxy cannot independently validate that same proxy; opposing province differences may cancel nationally. Coarse or fixed surfaces do not observe local trend changes. Satellite NO_2, SO_2, NH_3, AOD and fire products require transport, chemistry, retrieval sensitivity and sampling treatment. No simple column/concentration–emission correlation is treated as validation, and source factors are not tuned solely to match an external pattern.

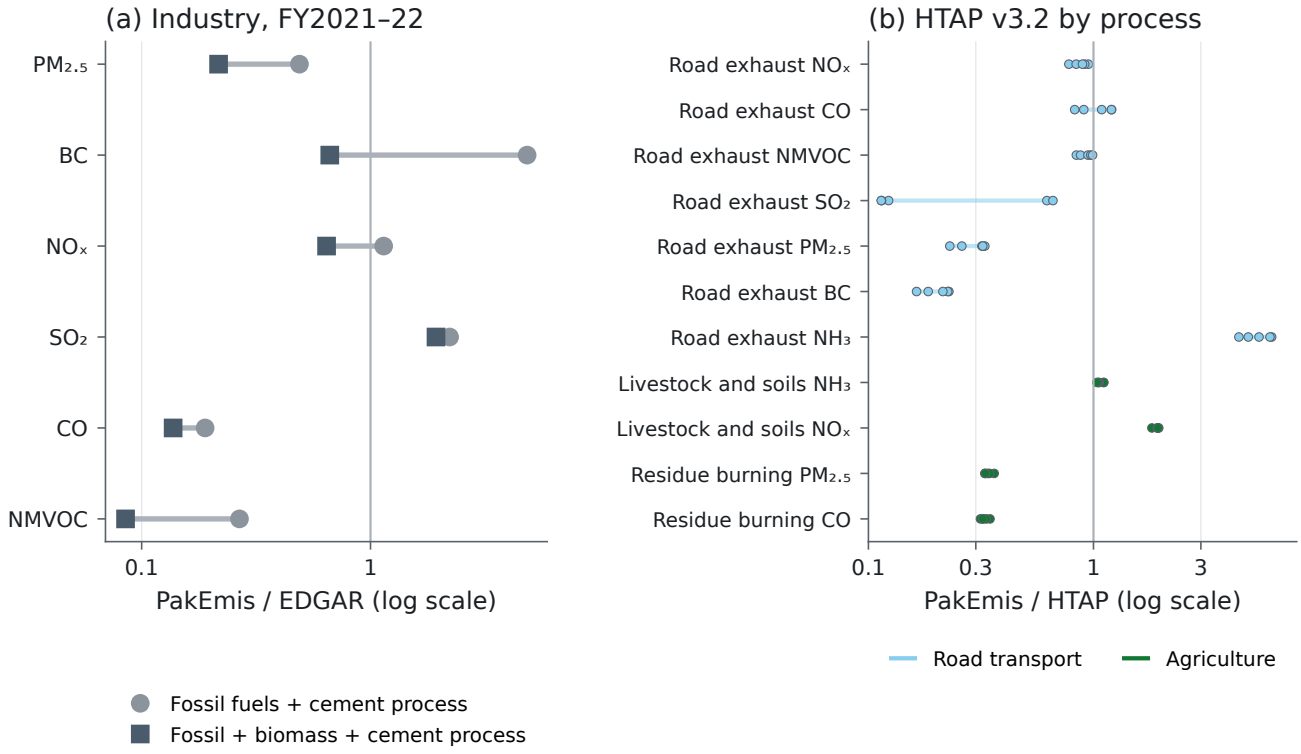


Figure 5. Air comparisons at documented fuel/process and exact fiscal-month boundaries. Industrial fossil/fossil-plus-biomass cases retain cement process matching; HTAP agriculture/road NH_3 uses published monthly records. Shared inputs limit independence.

6.4 Source importance and uncertainty priorities

Source contributions are evaluated within each native species or declared GWP metric. Agriculture dominates much of the methane/nitrogen account, while household fuel and open burning contribute substantial primary particulate. Industrial sulfur and combustion NO_x do not follow the same ordering. BC/OC and particulate strata are not summed to make a combined pollutant rank.

Source importance combines supported mass, plausible uncertainty and coverage. A small conditional percentile interval does not make an unestimated category negligible. The consequential scientific uncertainties include manure shares and soil-factor applicability for nitrogen, household dung destination and combustion factors for particulate, coal consumer allocation and fuel/control evidence for sulfur, and transferred road technology/class assumptions for NO_x.

Key-category and driver products retain the declared source/category/metric boundaries. Driver correlations are sensitivities rather than additive attribution. The main contribution is tracing which activity, factor, control, coverage or allocation can

explain an important difference, rather than listing a large number of small factors or ranking screening endpoints alongside central estimates.

7 Usage notes and limitations

Fitness for use depends on species, process, period, configuration and evidence support. Covered totals, conditional percentiles, structural cases, provisional sources and missing categories are distinct products. Administrative/cell allocations and monthly proxies must not be read as observed local emissions or atmospheric concentration.

7.1 Scope, period and source support

The common core spans ten fiscal years, July 2015–June 2025. It is not a full calendar-year 2015–2025 series. FY2023–24 remains the detailed reference uncertainty/spatial year because source support and latest annual availability answer different questions. FY2025–26 source/sector extensions are provisional and separate; their partial sums are not common-scope national totals.

A complete ten-module/year rectangle is compatible with substantial missing source categories. Native GHG and air products describe admitted sources under named configurations. External differences are not inventories of missing mass, and no unrelated global emissions are silently added. F-gases, coal-mining fugitives and remaining industrial/treatment sources retain explicit coverage records.

The scope is non-LULUCF. Exploratory land-sector estimates remain outside totals and are not treated as a validated correction to the national inventory. An omitted land flux may be a sink or source and cannot be interpreted as a one-direction bias.

7.2 Shared accounting and structural choices

Station coal consumption and rank-specific NCV define the power reference. Supply, receipts, stock change, consumer bookings and generation are distinct observations. Residual coal is explicitly unresolved where supported consumer evidence cannot allocate it; independently uncertain records do not become an exact physical balance merely because a published table is constructed as a residual identity. Dated kiln-fleet operating cases test credible alternatives without pretending to be an observed annual series or assigning them probabilities.

Household S uses survey dung and wood quantities, with historical interpolation or extrapolation labelled. L assumes all ledger-diverted dung is burned in households. The unobserved destination of remaining diverted dung is not resolved by material conservation. Survey recall, stove/fuel transfer factors and geographic proxies remain consequential uncertainty.

Fugitive methane remains conditional on practice, factor applicability and source coverage. Unaccounted-for gas includes non-leak components and is not equated to methane leakage. Native bounded source/fate draws are used where supported. Flaring and venting are distinct; their activity cannot be counted again in energy or IPPU.

7.3 Air methods and uncertainty

Air emissions use common activity but pollutant-specific factors, controls and chemical mass conventions. The seven core species have source–process coverage cells; industry, IPPU and fugitives include admitted provisional estimates. Power oil/diesel SO_2 is admitted separately from coal/gas/bagasse scenarios. Port Qasim and China Power Hub licences describe proposed design controls; installed or continuously operating removal is not demonstrated. Screening endpoints are typical-factor scenarios, not mathematical upper bounds.

Road methods include hot/cold exhaust, evaporation and tyre/brake/surface wear; resuspended dust and off-road air estimates remain absent. Transferred fleet-average factors, vehicle-age/technology composition, CNG-to-LPG chemistry and petrol-bus substitutions remain documented proxies. A valid fuel/CO_2 balance does not validate NO_x or particulate factors. Fuel-sulfur policy dates are specification evidence, not universal pump-level measurements.

Residential results are gross appliance releases. An outdoor escape fraction, vertical injection, detailed chemical speciation and atmospheric transport are not supplied. Waste burning geography follows origin rather than observed disposal/fire sites. Emissions rankings do not measure concentration, exposure or health-intervention priority. BC/OC and particulate strata are non-additive.

The TAN reference conserves stage-resolved nitrogen across NH_3, NO_x and N_2O; IPCC is a separate regenerated configuration. Conditional parameter ensembles share actual herd, crop, nutrient, waste and fugitive inputs and retain bounded fractions. Paired years produce trend intervals. Management shares, excretion, control operation, structural factor alternatives, missing categories and allocation uncertainty remain outside these percentiles. Dry-climate soil-factor applicability is evaluated separately for irrigated cropland, rice and grazing; matching FAOSTAT does not select a factor.

7.4 Timing and location

Monthly availability is assessed from the final exported activity, geographic coverage and compatible source controls. PDF group labels, continuation rows and worksheet identity are preserved when the PBS tables are read. Source anomalies that survive correct transcription remain flagged and an unsupported national profile is replaced by a declared weaker fallback. Twelve months alone never establish national observed activity.

Fertiliser offtake is a supply-timing proxy. Manure-stage, residue-decomposition, waste burning and unsupported evaporation/wear seasonality use declared fallback allocation. A CO_2-equivalent-weighted timing score does not describe an air pollutant; temporal and spatial evidence shares are calculated separately by species.

Grid cells carry allocated mass, not observed activity at $0.05°$. Fixed reference-year proxies do not establish changing local geography. Facilities respect commissioning/closure windows. Placed plus unallocated mass is conserved, and territories/-country masks retain their stated boundaries. Cell mass differs from flux density; area, country fraction and period duration are required for conversion.

7.5 Evaluation and availability

Matched comparisons separate magnitude, change, timing and location and record shared inputs. HTAPv3.2's direct country/monthly bundles are publicly available without credentials; a restricted optional spatial-subset service is not a restriction on those files. REAS/EDGAR inputs and EDGAR trend extensions limit independence. Satellite columns require chemistry, transport, retrieval and sampling treatment before they constrain emissions; a simple concentration–emission correlation is insufficient.

The review package contains current products, metadata, code, reproduction instructions and licences/checksums. Product-level licences distinguish independently sourced primary geography from ODbL-derived alternatives. A local package is not an external archive deposit, and no DOI or journal submission is asserted until that separately authorised action occurs.

7.6 Interpretation and contribution

The inventory's practical contribution is a Pakistan-specific source reconstruction that connects material and activity evidence to distinct GHG and air calculations. Station fuel and rank NCVs expose the public-power conversion choice. Industrial coal residuals and dated fleet alternatives expose allocation assumptions. Survey fuel distinguishes supported household combustion from other diverted material. Nitrogen stages explain linked NH_3, NO_x and N_2O consequences. These examples provide information that a clipped global inventory cannot supply from a national cell sum.

The common account does not make all pollutants equally assured. Reviewed oil/diesel SO_2 is admitted separately from coal/gas/control scenarios; industry/IPPU/fugitives include provisional source estimates; road evaporation and wear coexist with missing resuspended dust and off-road air sources. Species-specific coverage and controls therefore matter to every national comparison.

Household primary $PM_{2.5}$, agricultural NH_3, industrial sulfur and combustion NO_x describe different processes. There is no general inverse ordering of GHG and air rankings, and gross mass cannot rank exposure or health priorities. Missing vertical injection, chemical speciation and outdoor escape limits prevent an ambient-ready interpretation.

Annual extension and detailed reference choice are distinct. Later annual controls can support FY2025 magnitude without supporting new monthly release dates or finer source geography. The FY2026 provisional source products retain their exact periods and partial scope. Fixed location maps can distribute a changing national mass but do not establish changing local activity.

Shared conditional inputs and material closure improve coherence while leaving uncertainty in fate, controls, factor applicability and unestimated categories visible. Structural alternatives remain interpretable source cases without invented probabilities. The review package and consistent machine-readable records permit those decisions to be assessed and revised on evidence rather than hidden in one aggregate number.

8 Conclusions

PakEmis 1.0.0rc69 supplies one non-LULUCF greenhouse-gas and air-pollutant reconstruction for Pakistan over July 2015–June 2025. Native species and separate GWP metrics, source-process coverage, four integrated configurations, annual/monthly and geographic products, conditional uncertainty and source cases describe the same inventory account.

Station consumption/rank coal, reconciled consumer residuals, survey-supported household activity and conserved nitrogen pathways are consequential scientific changes. Supported FY2026 sources extend information provisionally without becoming a full national year. Public HTAP/EDGAR comparisons and source/export checks expose differences and limits rather than tuning estimates to agreement. Further improvement depends on source activity, management, factor applicability and demonstrated control evidence; nominal grid resolution alone cannot resolve those gaps.

All essential local review assets are prepared or identified by the release manifest and product-specific licensing record. A public repository review deposit and persistent dataset identifier remain separate external actions, with no submission or acceptance claim implied by this technical report.

9 Data availability

The dataset described here is PakEmis 1.0.0rc69. The dataset DOI is assigned on deposit; until then the release is identified by version and manifest digest (`outputs/release_manifest.json`, whose SHA-256 the release cover of every technical note prints). The technical documentation—the sector technical notes and this supplement—is included in the local release package. External review deposit and persistent identifiers are pending. The code is licensed under MIT. Project-owned primary products and text use CC BY 4.0; third-party inputs retain their stated terms. OpenStreetMap-derived databases and affected legacy spatial products are separated into an ODbL 1.0 companion. The primary geographic products are rebuilt without OpenStreetMap geometry or weights. The package includes the product-specific licence audit. One evaluation-only input carries no licence declaration in the record and is excluded from the primary data record until its rights are established (the manifest's `licences.undetermined` note). Product-specific licences and source custody are recorded in the release package. The primary geographic products exclude ODbL-derived effects at calculation time; preserved alternative database/geographic products retain their separate obligations. Source documents are named, located and hashed, without assuming that reading public information grants republication rights. A local review package is prepared; an external repository review deposit and persistent identifier remain pending.

10 Code availability

The software is release 1.0.0rc69 of the `pkinv` package (manifest schema `pkinv.release/v1`). The software DOI is assigned on deposit, with the dataset's; until then the release is identified by version and manifest digest. Source code is maintained at https://github.com/PakAirQuality/pakistan-emissions-sectors. This technical report describes PakEmis 1.0.0rc69, with exact code version, product hashes and reproduction level in the manifest/build record. Principal quantities are gener-

ated from current native data products. The national data-description article and maintained sector notes describe the same scope and configurations.

Author contributions. RA designed the inventory system, acquired and transcribed the source documents, wrote the engine and its tests, produced the data records and wrote the paper.

Competing interests. The author declares no competing interests.

Release provenance. This document describes PakEmis under the period, configuration and source boundaries specified in its methods. The identifiers below link it to the accompanying products and source register.

Release version	`1.0.0rc69`
Commit	`d8d874622f9a`
Reporting period	July 2015–June 2025 (ten fiscal years) (2015-07-01 to 2025-06-30)
Reference	S_TAN (household survey; TAN nitrogen)
FY2023–24 total	444.61 Mt CO_2e (AR5) under S_TAN, the reference; 449.85 Mt under L_TAN
Release manifest	`ac0ffe93ad9fcd88` (SHA-256 of `outputs/release_manifest.json`, first 16)
Source manifest	`c1be6d314cfce1ea` (SHA-256 of `data/manifests/source_manifest.csv`, first 16)
Data-description article	`paper_national/` (ESSD, in preparation), same release

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